

LoRD: LoRA-Adapted Re-Identification with DINO

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Abstract

We study video-based person re-identification on MARS using self-supervised DINO foundation backbones. Our approach ablates frozen DINOv2 and DINOv3 tracklet embeddings, learned mean pooling, learned temporal aggregation, and LoRA-adapted DINOv2 frame encoders under a common Re-ID training objective. We find that LoRA adaptation provides the dominant gain, raising MARS mAP from 1.8% zero-shot to 76.7% with a 2.6M-parameter LoRA-only model, while the addition of a learned temporal attention head fails to outperform simple mean pooling. Analysis of the trained temporal head shows that on LoRA-adapted DINOv2 features, the eight per-frame embeddings of one identity are already nearly collinear (cosine 0.97–0.99 in 1024-D), leaving little for cross-frame attention to exploit; training drives the head to a near-uniform pooling pattern (over 99% of maximum entropy on average across the query set). Loss and accuracy saturate by epochs 3–4, and final retrieval is approximately invariant to inference frame count between $T=4$ and $T=12$. Together, these results suggest that for LoRA-adapted DINO features, the video Re-ID bottleneck lies in per-frame representation quality rather than temporal reasoning—and that temporal attention may be largely redundant once the frame encoder is sufficiently adapted.

1. Introduction

Video-based person re-identification (Re-ID) matches a query tracklet (a short sequence of frames) against a gallery of tracklets across disjoint cameras, enabling cross-camera tracking in settings like retail surveillance, sports broadcasting, and traffic monitoring. Unlike image-based Re-ID, video Re-ID can exploit temporal context—motion, gait, and appearance variation—to resolve ambiguous identities, though doing so effectively remains an open problem. The naive implementation of a temporal transformer-based approach hurts performance without careful regularization

[17], while simple pooling discards the cues that distinguish video from image input.

Backbones for Re-ID are shifting from CNNs towards Vision Transformers [5] and, more recently, large foundation models. CLIP-based methods [15, 34] have strong semantic priors, but image–text contrastive pretraining is a poor proxy for identity matching. By contrast, DINO-family models [3, 24] address this mismatch via self-supervised objectives that produce locally discriminative features; however, their use in video Re-ID remains largely unexplored. Adaptation is another concern with Re-ID applications. Full fine-tuning of an 86M-parameter ViT-B is compute-intensive and risks forgetting the broader features that necessitate the use of a foundation model. LoRA [12] addresses this by injecting trainable low-rank residuals into a frozen backbone. Recent work suggests MLP-layer placement outperforms attention-projection placement at matched parameter budgets [27, 2], though this has not been validated on ViT backbones for Re-ID.

We present a systematic ablation study (see Figure 1) that isolates two design axes for video Re-ID on a DINO backbone: (i) the frame encoder (frozen versus LoRA-adapted) and (ii) the temporal aggregator (mean pooling, mean pooling with a learned linear projection, or a learned softmax-weighted temporal transformer head). All LoRA adapters are placed on the MLP feed-forward projections following recent evidence that this is the stronger placement [2, 27]. Our experiments on the MARS benchmark [35] allow us to attribute performance gains to specific components rather than to the overall system. We contribute to the broader literature by isolating the key factors that increase performance on video Re-ID tasks, and our main results are as follows.

- Frozen DINOv2 and DINOv3 zero-shot features are not sufficient for MARS video Re-ID, achieving 1.8% and 1.6% mAP respectively. This performance is consistent with published zero-shot frozen-foundation-model benchmarks on other Re-ID datasets [1].
- LoRA adaptation of DINOv2 (2.6M trainable param-

eters) delivers the largest gain in the ablation ladder, reaching 76.7% mAP and 80.6% Rank-1 on MARS within 10 epochs.

- A learned temporal transformer head with 12.6M additional trainable parameters does not improve over simple mean pooling on LoRA-adapted features. P5 (LoRA + temporal) underperforms P6 (LoRA + mean pool) by approximately 2 mAP across all tested inference frame counts.
- Training converges by epochs 3–4: classifier accuracy reaches 100% within the first few epochs while retrieval mAP plateaus, indicating that gains come from backbone adaptation rather than head capacity.
- Final retrieval performance is approximately invariant to inference frame count: P5 and P6 results at $T=4$, $T=8$, and $T=12$ differ by less than 1 mAP, suggesting that the number of provided frames at inference does little to benefit an already-trained backbone.

2. Related Work

Person re-identification has a long history as a computer vision problem, evolving from hand-crafted feature descriptors and metric learning approaches [16, 36] to deep convolutional networks. Early deep Re-ID systems learned discriminative appearance embeddings using identity cross-entropy and pairwise metric losses [10], establishing the triplet loss strategies that remain standard today. For video Re-ID specifically, the best-performing methods paired CNN frame encoders with recurrent or 3D-convolutional temporal modules to aggregate appearance over time [22, 14]. AP3D [8] introduced appearance-preserving 3D convolutions that avoid temporal blurring and reached competitive accuracy while keeping the backbone lightweight. These CNN-era models established the core evaluation protocols of mAP and CMC Rank-1 on MARS [35] that we follow throughout this study.

2.1. ViT-Based Re-ID

The transition to transformer architectures began with evidence that self-attention over frame tokens could match 3D-CNN performance on video Re-ID. TransReID [9] then established the canonical ViT Re-ID training recipe for the image setting: overlapping patch embeddings, a camera-ID side-embedding, and a jigsaw patch regularizer, evaluated with BNNeck [18]. This pipeline, extended to self-supervised backbones in TransReID-SSL [19], serves as our primary training scaffold. KeyRe-ID [13] combines global temporal attention with keypoint-guided local part pooling, achieving state-of-the-art accuracy on MARS and LS-VID and serving as our primary ViT video Re-ID comparison.

Foundation models have recently entered this space. CLIP-ReID [15] proposes a two-stage strategy using learnable text tokens to bridge visual and language identity features. TF-CLIP [34] removes the text-supervision requirement with CLIP-Memory and Temporal Memory Diffusion (TMD), making it the most directly comparable foundation-model baseline for text-free video Re-ID. On the self-supervised side, DINO [3] and DINOv2 [24] produce dense, locally discriminative features that transfer well to matching tasks, and DINOv2 with register tokens [4] further stabilizes the dense feature quality. A recent survey [1] benchmarks frozen DINOv2 on nine Re-ID datasets (mAP 0.3–4.7), confirming that some form of adaptation is necessary. We extend this line to video Re-ID, and include DINOv3 [29] as a stretch comparison, finding that under matched model size it surprisingly underperforms DINOv2 on MARS in a zero-shot setting.

2.2. Training Methods for ViT Re-ID

The quality of Re-ID embeddings depends heavily on the loss landscape and training recipe. The most consequential single contribution in this regard is BNNeck [18]: a batch normalization layer inserted between the backbone embedding and the ID classifier. The key insight is that cosine-distance triplet loss and softmax ID loss prefer different embedding geometries—BNNeck decouples them by applying the ID head to the post-BN feature while retaining the pre-BN feature for triplet loss and inference. We adopt this training methodology for our study. For parameter-efficient adaptation, LoRA [12] offers a principled way to fine-tune large ViTs by injecting low-rank residuals into linear layers while keeping the backbone frozen; recent work [27, 2] recommends placing these residuals in the MLP (FFN) layers rather than the attention projections, which we implement.

2.3. Re-ID Datasets

Re-ID benchmarks span a wide range of settings and difficulty levels. For standard video Re-ID, MARS [35] (1,191 identities, 6 cameras, automatically generated tracklets) is the primary benchmark and our main evaluation target. Its noisy, detector-generated tracklets make it a realistic proxy for real-world deployment. LS-VID [14] scales this up to 3,772 training identities across 15 cameras with more severe outdoor illumination variation; we do not evaluate on LS-VID in this work, but it could be used for generalizability testing in future extensions. For image-based Re-ID, Market-1501 [36] and DukeMTMC-reID [26] are the canonical benchmarks, and strong results there motivated the ViT architectures we build on. Specialty datasets address particular deployment challenges: Occluded-DukeMTMC [23] and P-DukeMTMC [20] target partial and occluded pedestrians; MSMT17 [32] covers large-scale multi-scene variation; and iLIDS-VID [30] and

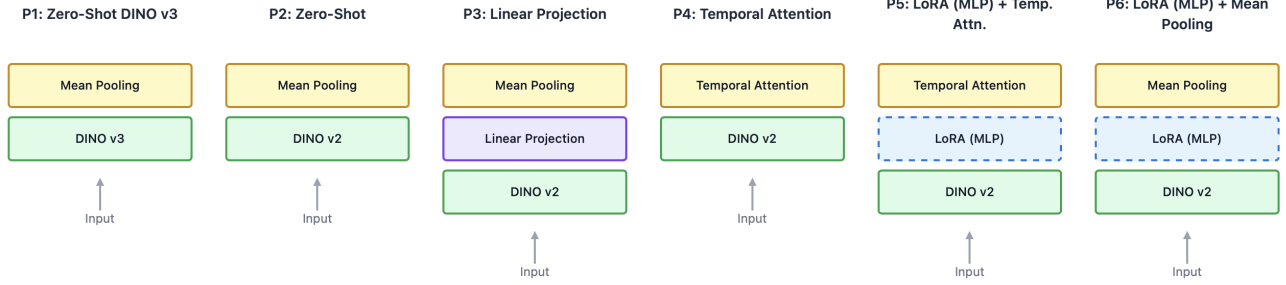


Figure 1. **Ablation Ladder.** We structure our experiments as shown to isolate the role of each component in improving Re-ID performance.

PRID-2011 [11] are smaller-scale video benchmarks used for low-data generalization testing.

3. Data

MARS (Motion Analysis and Re-identification Set) [35] is our primary benchmark for video-based person Re-ID. It contains approximately 1,191 identities captured across six cameras in a university campus setting, with tracklets automatically generated by a DPM detector and GMMCP tracker. The standard split allocates 625 identities for training (8,298 tracklets) and 626 identities for testing (1,980 query + 9,330 gallery tracklets, including 3,248 distractor tracklets). Automatic tracklet generation creates labeling noise—estimated at 3.9% error—and frequent occlusions, illumination changes, and viewpoint variation, making MARS the de facto evaluation for much of the Re-ID literature referenced in this study.

We perform the following train and test-time data augmentations. To choose frames, we randomly sample $T=4$ frames per tracklet during training and $T=8$ at inference per the published MARS protocol, with additional $T=4$ and $T=12$ evaluations to test sensitivity to inference frame count. We conduct a stratified sample such that each temporal segment is equally represented in the final sample. We also use PK sampling with $P=16$ identities and $K=4$ tracklets per batch, and the final LoRA runs use $P=8$, $K=4$ for memory efficiency. We apply tracklet-consistent data augmentations during training that include: random padded crops (pad 10), random horizontal flips (probability 0.5), and random erasing (probability 0.5, area in $[0.02, 0.4]$). In our pipeline, each frame runs independently through the ViT backbone—the primary source of comparison is the ways in which we combine the embeddings post-backbone. This approach is detailed in Figure 2 and the following section.

4. Methods

Our implementation is organized as an ablation ladder. This lets us isolate whether performance comes from the

changes to the DINO backbone (frozen or LoRA) or to the aggregation method (attention vs. pooling). All models consume a sampled tracklet $x \in \mathbb{R}^{T \times 3 \times H \times W}$ and produce one tracklet-level embedding for retrieval.

4.1. Ablation Paradigms

Figure 1 summarizes the six implemented paradigms. P1 and P2 are zero-shot frozen-backbone baselines aimed at finding the best backbone for the rest of our experiments. P3–P6 are trained with the same Re-ID objective and differ only in adaptation and temporal aggregation. We used a personal NVIDIA GeForce RTX 5070 GPU for training.

4.2. Frame Encoder

The frame encoder for all experiments is a DINO ViT loaded with the model’s standard image processor from HuggingFace. For DINOv2 experiments, we use the register-token ViT-L/14 checkpoint `facebook/dinov2-with-registers-large` [24, 4]; for the DINOv3 zero-shot evaluation (P1), we evaluate DINOv3 ViT-L/16 [29]. Each frame is resized to 252×126 , preserving the approximately 2:1 Re-ID pedestrian aspect ratio while aligning both dimensions to the ViT-L/14 patch stride. We use the CLS token as the frame descriptor and do not pool register tokens into the retrieval embedding.

For LoRA experiments (P5 and P6), the DINO attention weights remain frozen and PEFT LoRA adapters are inserted into the DINOv2 MLP layers [12]. Our default configuration uses rank $r=8$, $\alpha=16$, and dropout 0.05. This choice follows recent evidence that MLP-targeted LoRA can be a stronger and less forgetting-prone adaptation path than the original Q/V-only convention [2, 27].

4.3. Temporal Aggregation

The mean-pooling experiments (P1 and P2) simply average the T frame CLS embeddings:

$$z = \frac{1}{T} \sum_{t=1}^T f_{\theta}(x_t). \quad (1)$$

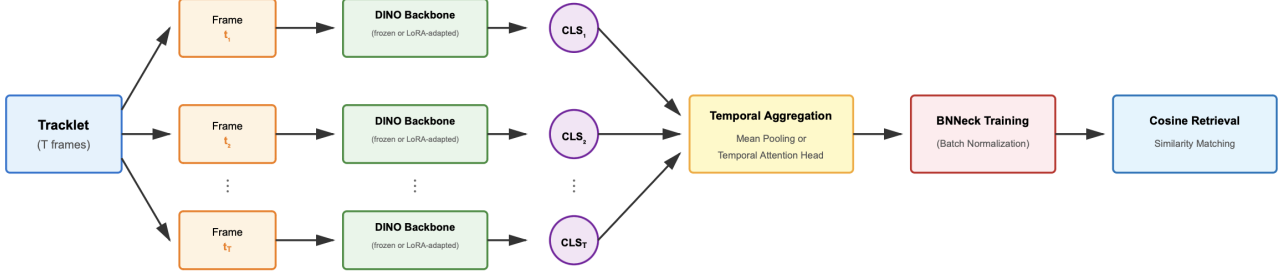


Figure 2. **LoRD architecture overview.** Each tracklet is sampled into T frames, each frame is encoded independently by a frozen or LoRA-adapted DINO backbone, and the resulting frame-CLS tokens are aggregated by either mean pooling or a lightweight temporal transformer before BNNeck training and cosine retrieval.

P3 adds a learned linear projection after this mean. The temporal-attention experiments (P4 and P5) instead add a sinusoidal time embedding to the frame CLS sequence and pass it through a transformer encoder. Our final temporal head uses one pre-norm transformer block with multi-head self-attention, a GELU MLP, dropout, and residual connections, followed by a learned softmax over frames:

$$\alpha_t = \text{softmax}(w^\top h_t), \quad z = \sum_{t=1}^T \alpha_t h_t. \quad (2)$$

The attention output projection, final temporal-MLP projection, temporal pooling logits, and positional scale are initialized at zero so that the module starts close to mean pooling and learns cross-frame mixing only when useful. This design keeps temporal modeling lightweight and tests whether a KeyRe-ID style frame-weighting mechanism [13] improves over simpler pooling on top of DINO features.

4.4. Training Objective

For P3–P6, the tracklet embedding is trained with the standard Re-ID combination of identity classification and metric learning [18, 10]. A BNNeck layer maps the pre-BN embedding z into a post-BN feature used by the identity classifier. We apply label-smoothed cross-entropy with smoothing 0.1 to classifier logits and batch-hard triplet loss with margin 0.3 to the pre-BN embeddings:

$$\mathcal{L} = \mathcal{L}_{CE} + \mathcal{L}_{triplet}. \quad (3)$$

At inference, we discard the classifier and use the L2-normalized post-BN embedding for cosine retrieval.

4.5. Optimization and Sampling

Training uses PK sampling with P identities and K tracklets per identity so that every batch contains positives and hard negatives for triplet mining. The default implementation supports $P=16, K=4$, but our final reported LoRA runs use $P=8, K=4$ for memory efficiency. During

training, each tracklet is split into T contiguous segments and one frame is sampled from each segment. At evaluation, frames are selected deterministically at approximately uniform positions, and short tracklets are padded to reach the minimum frame count.

We train with AdamW, using separate learning rates for adapter and head parameters (5×10^{-5} for LoRA and 3.5×10^{-4} for the temporal head, projection, BNNeck, and classifier). We also use weight decay of 10^{-4} except on normalization layers, BNNeck, the temporal pool logits, and the positional-scale parameter, gradient clipping at 1.0, a linear warmup of 5 epochs for P3 and P4 and 1 epoch for P5 and P6, and a cosine decay. The final LoRA runs (P5 and P6) were configured for 7 epochs with $T=4$ training frames, using fixed random seed 42 to ensure both paradigms saw identical PK batches and augmentation patterns. While the published MARS protocol evaluates at $T=8$, we also evaluate at $T=4$ and $T=12$ to test whether providing more frames at inference benefits any specific aggregation design (mean pooling versus temporal attention).

4.6. Evaluation

We evaluate on the official MARS query/gallery split. Query and gallery tracklets are embedded once, pairwise cosine distance is computed, and results are ranked per query. Following the official MARS protocol, gallery tracklets with distractor identity -1 and same-identity same-camera matches are treated as junk and removed for each query before computing CMC and AP. We report mAP and CMC Rank-1 as headline metrics, with Rank-5 and Rank-10 used for secondary analysis.

5. Experiments

5.1. Ablation Results

We train and evaluate DINO-based models across the 6 paradigms. We report the mAP and CMC across the six paradigms as outlined in figure 1 and table 1. The frozen

ID	Trainable	T	mAP	R1	R5
P1	0	8	0.0164	0.0384	0.0763
P2	0	8	0.0181	0.0364	0.0798
P3	1.7M	8	0.2999	0.4045	0.6212
P4	13.3M	4	0.2166	0.3167	0.5207
P5	15.2M	4	0.7447	0.7828	0.8955
P5	15.2M	8	0.7511	0.7874	0.8949
P6	2.6M	4	0.7586	0.7990	0.8970
P6	2.6M	8	0.7660	0.8030	0.8975

Table 1. **P1–P6 MARS results.** P5 and P6 are the LoRA paradigms reported at the last completed epoch (epoch 7). P3 and P4 are 60-epoch frozen-backbone runs. Trainable parameter counts exclude the frozen DINOv2 backbone (~ 304 M).

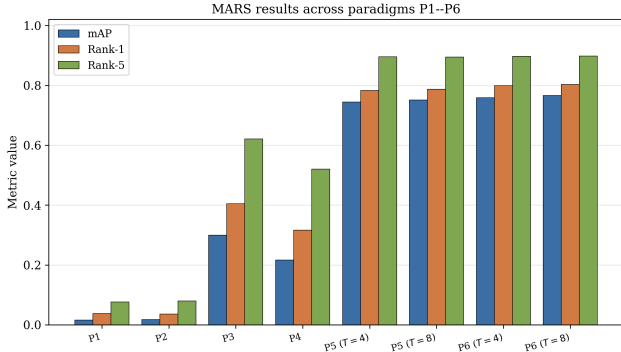


Figure 3. **P1–P6 MARS results, visualized.** The same data as Table 1, shown as grouped bars.

zero-shot baselines (P1 and P2) are not sufficient (1.6%–1.8% mAP), consistent with the 0.4%–4.7% range found in the literature. A frozen DINOv2 backbone with a mean-pool + learned linear projection (P3) reaches 30.0% mAP / 40.5% Rank-1. The frozen temporal transformer (P4) underperforms this projection baseline (21.7% mAP at $T=4$), indicating that temporal attention does not pay off when the frame encoder is fixed (addressed below). LoRA adaptation is the dominant paradigm: P5 (LoRA + temporal, 15.2M trainable parameters) reaches 74.5% mAP / 78.3% Rank-1, and P6 (LoRA + mean pool, 2.6M trainable parameters) reaches 76.7% mAP / 80.6% Rank-1, outperforming P5 with $\sim 6\times$ fewer trainable parameters.

5.2. Learning Dynamics

The final P5 and P6 runs show that most of the gain comes within the first few epochs (training-loss and validation curves are in the Appendix, Figures 8 and 9). P5 improved from 62.0% mAP at epoch 1 to 71.4% by epoch 3 and 75.1% by epoch 6, while P6 improved from 58.1% mAP at epoch 1 to 72.7% by epoch 3 and peaked near 76.7% at epochs 5–6. Classifier accuracy on the training PK batches saturates at 100% within the first few epochs while retrieval mAP plateaus, indicating that the additional

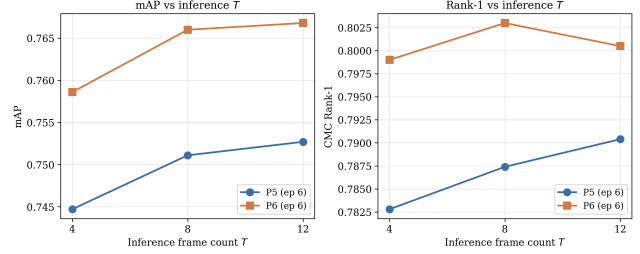


Figure 4. **Frame-count sensitivity.** mAP and Rank-1 at $T=4, 8, 12$ for the final P5 and P6 checkpoints. Mean pooling and temporal attention both show small monotonic gains with more frames; the gap between paradigms is preserved across all T .

training epochs do not contribute performance gains once the LoRA-adapted backbone has converged.

5.3. Temporal Aggregation Comparison

The conclusion from the LoRA runs is that backbone adaptation provides the dominant gain, but the temporal transformer head does not justify its added complexity. P6 (LoRA + mean pooling, 2.6M trainable parameters) reaches higher mAP and Rank-1 than P5 (LoRA + temporal attention, 15.2M trainable parameters) at every inference frame count we tested. A single temporal transformer head adds 12.6M trainable parameters (approximately $6\times$ more than P6’s full trainable budget) without improving retrieval. Mechanistic analysis of the trained P5 attention head (Section 5.5) shows the head converges to near-uniform pooling on most tracklets, and the residual structure ensures that it contributes only a small perturbation to the final embedding. This does not rule out temporal attention in situations with increased frame volume or with a broader hyperparameter sweep, but it argues against treating temporal attention as the primary driver of performance in this setting.

5.4. Frame-Count Sensitivity

Figure 4 reports mAP and Rank-1 at $T=4, T=8$, and $T=12$ for the final P5 and P6 checkpoints. Both paradigms exhibit small monotonic improvements with increasing T , and the gap between them is preserved across all three values. The total spread across T is under 1 mAP for either model, showing that final retrieval performance is approximately invariant to inference frame count once the backbone is adapted. The temporal head (P5) does not extract proportionally more value from additional frames than mean pooling (P6) does, consistent with the analysis in Section 5.5.

5.5. Why Temporal Attention Does Not Help

To understand why the temporal head fails to improve over mean pooling, we analyze the trained P5 model and examine its attention behavior on the MARS query set

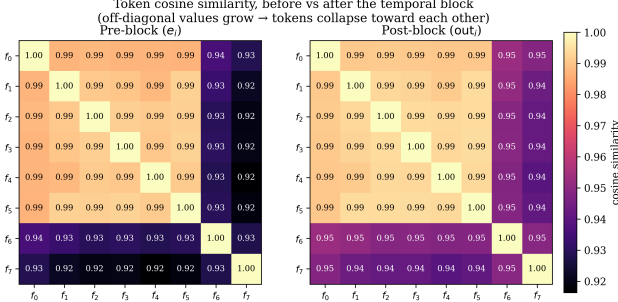


Figure 5. **Per-frame token cosine similarity, before vs after the temporal block** (query tracklet #426).

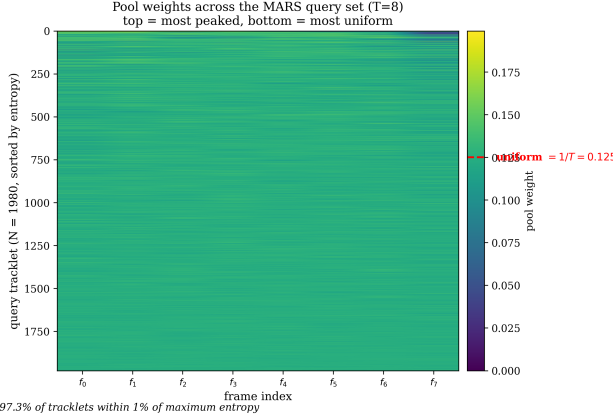


Figure 6. **Final attention-pool weights across the MARS query set** (1,980 tracklets, $T=8$), sorted by entropy. The bulk of tracklets show pool weights within 1% of fully uniform.

(1,980 tracklets, $T=8$). Three results pinpoint the underlying cause.

Frame embeddings already have high cosine similarity. Figure 5 shows the pairwise cosine similarity of the eight DINOv2 frame CLS tokens for one query tracklet, before and after the temporal block. Before the temporal layer, the similarities are 0.92–0.99, and the transformer block nudges these values upward by at most 0.02. With per-frame embeddings this clustered any reasonable pooling (mean, attention, single-frame selection) produces nearly the same combined embedding.

The learned head converges to near-uniform pooling. Across all 1,980 query tracklets, the final attention-pool entropy is on average 99.7% of the theoretical maximum ($\log T$). The single most-peaked tracklet in the dataset reaches only 96% of maximum entropy. Figure 6 visualizes the per-tracklet pool weights sorted by entropy: the bulk of the dataset shows nearly uniform pool weights, and even the highest entropy tracklets still are relatively uniform.

Per-head attention shows selection but residual structure washes it out. Figure 7 shows the attention pattern for the single most-peaked query tracklet. Six of eight

self-attention heads converge to a “broadcast” pattern in which every query position attends to one “most informative” frame (f_0). However, the architecture zero-initializes the weight of the self attention output and it doesn’t grow to be nearly as significant as the residual path, so the per-frame DINOv2 embedding is preserved through. Thus, the selected “most informative” signal contributes only a small correction term on top of the original frame embeddings, and, when combined with near-uniform pool weights, produces an output approximately equal to mean pooling. Additionally, the temporal head adds sinusoidal positional encodings (which mark which frame is first, second, third, and so on) to the frame embeddings before self-attention, scaled by a learnable scalar pos_scale . This scalar is initialized to zero and grows during training only if frame order helps reduce the loss. At convergence we measure $\text{pos_scale} = 0.018$, which amounts to approximately 2% of the natural magnitude of the positional encoding vectors and means that the trained head effectively ignores frame order and treats the tracklet as an unordered set of frames.

We interpret this as an instance of a more general phenomenon: ensembling beats selection when per-frame features are already highly discriminative. On LoRA-adapted DINOv2 features, the eight per-frame embeddings of one identity are already concentrated together, so mean pooling them produces approximately the same direction as any weighted combination would, and the additional capacity of the temporal transformer provides no meaningful performance improvement.

5.6. Published MARS Comparison Anchors

Our reported metrics use direct cosine retrieval only. This makes the comparison to published MARS systems conservative because such systems use additional post-processing. In particular, the k-reciprocal re-ranking method of Zhong et al. [37] combines the original distance with a Jaccard distance over reciprocal nearest neighbors and requires no additional labels. In a controlled MARS study with a ResNet-50 frame encoder, Gao and Nevatia [7] report that their best temporal-attention model improves from 76.7 mAP / 83.3 Rank-1 to 84.5 mAP / 85.0 Rank-1 after re-ranking, a +7.8 mAP gain. We do not include re-ranking in this submission, as it is best treated as a future post-processing extension rather than part of the present experimental setup. Table 2 therefore uses a small set of models chosen to separate four questions: how our direct-retrieval result compares to an older CNN temporal baseline, how much post-processing can change MARS numbers, where 3D CNNs sit, and how far current CLIP/ViT video systems have pushed the benchmark. It should be noted that our P6 model achieves competitive accuracy when compared to the selected models we present in Table 2 despite the objective of this study being not to achieve

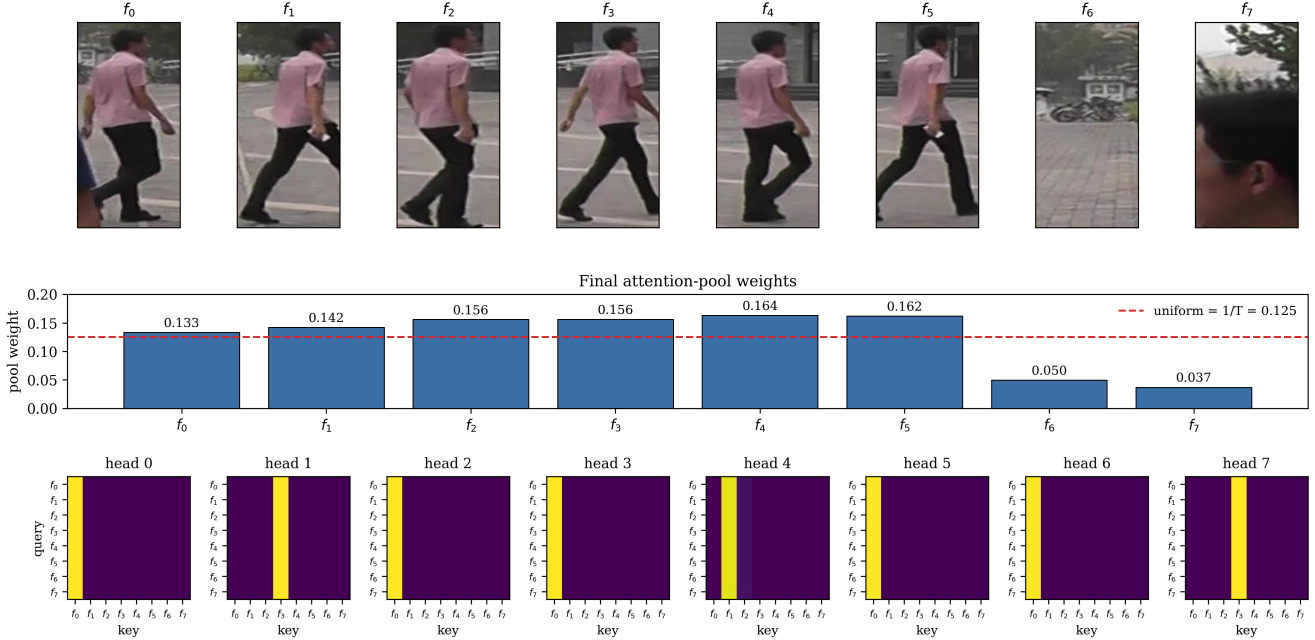


Figure 7. **Trained P5 attention pattern** on the single most-peaked query tracklet (#426). Top row: the eight input frames. Middle: final attention-pool weights with uniform reference; the model downweights the two off-target frames (f_6, f_7) but the deviation from uniform is mild. Bottom: per-head self-attention matrices, where rows are queries and columns are keys. Six of eight heads converge to “broadcast frame f_0 ” (vertical stripe). The residual connection preserves each frame’s DINOv2 embedding, so this selection contributes only a small correction term to the final tracklet vector.

Method	Family	R1	mAP
Gao+Nevatia att.	CNN+temporal	83.3	76.7
Gao+Nevatia rerank	CNN+rerank	85.0	84.5
AP3D	3D CNN	90.1	85.1
TF-CLIP	CLIP video	93.0	89.4
KeyRe-ID	ViT video	97.3	91.7
LoRD (Ours) P6, $T=8$	LoRA-DINO mean	80.6	76.7

Table 2. **MARS comparison anchors.** Published rows are not reimplemented here and may use different training data, backbones, or re-ranking settings.

SOTA but rather to probe ablations. This shows promise for future DINO-based Re-ID models.

5.7. Possible Future Ablations

Because the final runs use one LoRA configuration ($r=8$, $\alpha=16$, dropout 0.05) and a limited temporal-head setting, our conclusion is framed conservatively. Important follow-up axes are LoRA rank ($r \in \{4, 8, 16\}$), LoRA placement (MLP versus Q/V versus both), temporal dropout, number of temporal layers/heads, training frame count, and feature extraction from the final hidden layer rather than only the CLS token. Optimizing hyperparameters for this objective could yield performance gains that we were unable to capture and should be explored. We also believe that a large reason that temporal attention was unable

to elicit performance gains was because of the low frame count in the MARS dataset coupled with the lack of variation between images. Though a longer time-scale dataset is not easily accessible, future work in that area could definitively conclude the efficacy of temporal attention layers in Re-ID. Separately, k-reciprocal re-ranking is a promising post-processing direction, but we leave it outside the scope of the current submission.

6. Conclusion

Key result. Adaptation of the DINOv2 frame encoder is the dominant factor in this system. Frozen DINO features provide a useful starting point but do not solve MARS zero-shot (1.6%–1.8% mAP); LoRA on the MLP feed-forward layers moves retrieval into a much stronger regime (76.7% mAP for P6) within only a few epochs while training 2.6M parameters, compared with ~ 25.6 M parameters for a fully trainable ResNet-50 baseline. This places parameter-efficient adaptation of self-supervised ViT foundation models as a competitive approach for video Re-ID, even before considering longer training, re-ranking, or larger-scale data.

Temporal result. The temporal transformer head does not currently justify its additional parameters. Both the frozen temporal model (P4) and the LoRA temporal model (P5) underperform their simpler mean-pooling counterparts (P3 and P6 respectively). Mechanistic analysis of the

trained P5 attention shows the head converges to near-uniform pooling (over 99% of maximum entropy across the query set) and learns a positional gate $\text{pos_scale} = 0.018$ that effectively ignores frame order. The underlying cause is that LoRA-adapted DINOv2 produces frame embeddings of the same identity that are already nearly collinear (cosine $\sim 0.97\text{--}0.99$ in 1024-D), leaving little structure for cross-frame attention to exploit. Final retrieval is therefore approximately invariant to inference frame count between $T=4$ and $T=12$.

Caveat. This is an implementation-scale conclusion, not a claim that temporal reasoning is irrelevant to video Re-ID. Temporal attention may matter with weaker per-frame backbones (where frame diversity is larger), longer or noisier tracklets where some frames are uninformative, larger datasets, stronger regularization, patch-level temporal cues rather than CLS-only tokens, or a broader hyperparameter search.

Future work. A natural follow-up is to sweep LoRA rank ($r \in \{4, 8, 16\}$) and placement (MLP vs. Q/V vs. full) on the same training pipeline. We would also tune temporal head depth, heads, and dropout, and test last-layer patch-token or multi-token pooling instead of CLS-only features following [28, 31]. K-reciprocal re-ranking [37] is a low-cost post-processing direction we did not pursue here, but published work suggest that it can increase mAP. Cross-benchmark generalization on LS-VID and MEVID would test whether the LoRA-adapted DINOv2 baseline transfers, and whether temporal aggregation becomes more important on those harder splits. Additionally, we hypothesize that the generalizability of a ViT backbone would increase performance on occluded video scenarios, but we were unable to increase the scope of this project to evaluate on such material. We would like to evaluate our P5 and P6 trained models on such datasets and analyze the results to test this hypothesis.

7. Contributions and Acknowledgments

Author contributions. All authors contributed equally to this project. Joseph focused on training architecture design and familiarization with existing Re-ID literature in addition to implementing evaluation and plotting infrastructure. Anand focused on implementation of the training infrastructure and data processing code. Andres focused on running the experiments and debugging train-time errors. However, all authors were involved with all aspects of the project. All authors contributed to writing the final report and poster.

Code and reproducibility. Our training and evaluation pipeline is available at <https://github.com/anandkrishnan27/latrine>. We build on Hugging Face Transformers [33] for the DINOv2/DINOv3 backbone loaders, PEFT [21] for LoRA

adapter insertion, and PyTorch [25] with PyTorch Lightning [6] for the training loop. Our MARS evaluation script is a Python re-implementation of the official MATLAB evaluator (`evaluation_mars.m`, `compute_AP.m`) released with [35]; we preserve the official junk-filtering and trapezoidal-AP computation. No pretrained Re-ID checkpoints from other published systems were used.

Compute. All training and evaluation runs were executed on a single NVIDIA GeForce RTX 5070 (12 GB VRAM) belonging to Andres. The 60-epoch frozen-backbone runs (P3, P4) took approximately 1.5–2 hours each; the 10-epoch LoRA runs (P5, P6) took approximately 9–10 hours.

Class scope. This work is submitted for CS231N only and is not shared with another class.

Generative AI Usage Statement

We used Anthropic’s Claude (Claude Code interface) as a coding assistant throughout the project. Specifically, the tool was used for: (1) debugging PyTorch and PyTorch Lightning errors during dataset and model implementation; (2) generating boilerplate for argument parsing, dataloader configuration, evaluation scripts, and matplotlib figure code; (3) drafting and refining the visualization scripts that produced the attention analysis figures (Figures 7, 5, and 6); and (4) reformatting the LaTeX manuscript to adhere to LaTeX syntax.

All design decisions for the ablation paradigms (P1–P6), the training configuration (PK sampling, BNNeck, label-smoothed CE + batch-hard triplet, LoRA placement on the MLP), the choice to test $T=4/8/12$ inference sensitivity, and the interpretation of all empirical results were made by the team. The AI tool did not generate the numerical results or run experiments without supervision; all metrics in this paper come from runs we explicitly launched and monitored.

8. Appendix

Supplementary training and validation curves for the final P5 and P6 LoRA runs are included here for reference. They support the claims in Section 5 that retrieval performance saturates by epochs 3–4 and that P6 (LoRA + mean pool) leads P5 (LoRA + temporal) by 1–2 mAP throughout training.

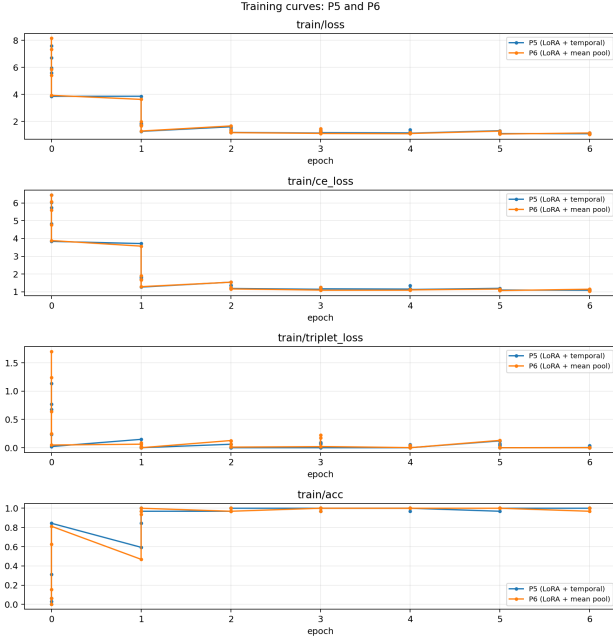


Figure 8. **Training curves for P5 and P6.** Combined loss, cross-entropy, batch-hard triplet, and classifier accuracy across the 7 epoch runs. Both paradigms reach 100% PK-batch classifier accuracy within the first few epochs while losses continue to decrease modestly.

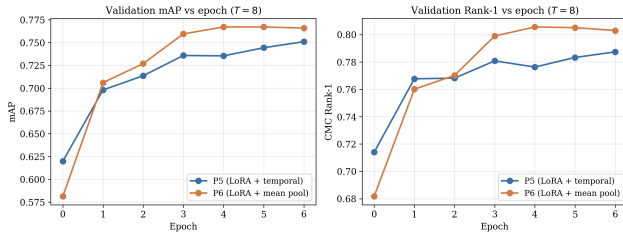


Figure 9. **Validation mAP and Rank-1 vs epoch** for P5 (LoRA + temporal) and P6 (LoRA + mean pool) at $T=8$ inference. P6 leads P5 by roughly 1–2 mAP across all late epochs.

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